

Chapter #09 (Review of Vectors)

(01)

Q.02

Definition (Scalar quantity)

A scalar is a quantity that is determined by its magnitude. It takes on a numerical value i.e., numbers.

Ex Time, length

Vector quantity

Vector is a quantity that both has magnitude and direction.

Ex Velocity, Force

Remark

A vector of length 1 is unit vector.

Length

$$|a| = \sqrt{a_1^2 + a_2^2 + a_3^2} \quad (1)$$

Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ In $\mathbb{R}^2$,

$$\vec{a} = [a_1, a_2] = (a_1, a_2) = a_1 \hat{i} + a_2 \hat{j} \quad (2)$$

$\hat{i} = (1, 0)$
 $\hat{j} = (0, 1)$

in $\mathbb{R}^3$,

$$\vec{a} = [a_1, a_2, a_3] = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \quad (3)$$

$\hat{i} = (1, 0, 0)$
 $\hat{j} = (0, 1, 0)$
 $\hat{k} = (0, 0, 1)$

Inner / Dot Product of two vectors

$$\text{Let } \vec{a} = [a_1, a_2, a_3] = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \quad (4)$$

$$\vec{b} = [b_1, b_2, b_3] = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k} \quad (5)$$

So,

$$\vec{a} \cdot \vec{b} = (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \cdot (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k})$$

$$= a_1 b_1 + a_2 b_2 + a_3 b_3$$

$\hat{i} \cdot \hat{i} = 1 = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k}$ $\hat{i} \cdot \hat{j} = \hat{k} \cdot \hat{i} = \hat{j} \cdot \hat{k} = 0$
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Remark

q¹ The inner product of two non-zero vectors is 0 iff these vectors are perpendicular.

q² angle b/w two vectors:

$$\cos \alpha = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \quad (1)$$

* (Application of Dot product) is work done

* work done by force is expressed by inner product.

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Cross product (vector product)

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

then,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \hat{i}(a_2b_3 - a_3b_2) - \hat{j}(a_1b_3 - b_1a_3) + \hat{k}(a_1b_2 - a_2b_1)$$

Ex

Find vector product of

$$\vec{a} = [1, 1, 0], \vec{b} = [3, 0, 0]$$

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Sol

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 3 & 0 & 0 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 0 \\ 3 & 0 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 1 \\ 3 & 0 \end{vmatrix}$$

$$= \hat{i} (0 - 0) - \hat{j} (0 - 0) + \hat{k} (0 - 3)$$

$$= -3\hat{k} = 0\hat{i} + 0\hat{j} - 3\hat{k}$$

$$\boxed{\vec{a} \times \vec{b} = [0, 0, -3]}$$

Scalar function

A scalar function is a function whose result or output is scalar quantity.

$$\boxed{\phi = \phi(x, y, z)} \quad \text{--- (*)}$$

$$\phi(x, y, z) = x^2 \sin(yz)$$

$$\boxed{\phi(0, 0, 0) = 0}$$

Vector Field (Introduction)

(is a systems of vectors)

Generally, a vector field on a domain in the plane or in space is a function that assigns a vector to each point in the domain.

A field of two-dimensional vector is,

$$\boxed{F(x, y) = M(x, y) \hat{i} + N(x, y) \hat{j}} \quad \text{--- (1)}$$

A field of three-dimensional vector is

$$\boxed{F(x, y, z) = M(x, y, z) \hat{i} + N(x, y, z) \hat{j} + P(x, y, z) \hat{k}}$$

--- (2)